Frontier Field-Dynamics in Action:

Structural Impossibility Results under Fixed Observational Granularity

Fabrizio De Palma
Independent Researcher

Abstract

This paper investigates what happens when **Frontier Field-Dynamics (FFD)** is applied to concrete mathematical settings in theoretical computer science and adjacent areas.
FFD, introduced in a companion foundational work, is a structural framework for analyzing reasoning and search processes under **fixed observational granularity**, identifying regions—frontiers—where internal evolution yields no observable informational progress.

We show that applying FFD does not produce isolated phenomena, but systematically forces **structural impossibility results**.
In particular, we analyze two distinct frontier mechanisms.

First, we prove a **finite-grain non-factorization theorem for the counting problem #SAT**: for any finite observational scheme, the counting function fails to stabilize even pointwise. This result is model-independent and reveals an intrinsic oscillation phenomenon that prevents localization of global counts from finite information.

Second, we establish a **general no-go theorem for canonical selection under natural coherence constraints**, showing that no non-trivial selection rule can be simultaneously intrinsic, compatible with loss of information, and stable under extension. This obstruction is purely structural and yields, as immediate corollaries, familiar non-canonicity phenomena in general topology (absence of canonical minimal bases), Ramsey-type settings (impossibility of equivariant witness selection), and concurrent systems such as Petri nets.

These results are not presented as independent discoveries, but as **inevitable manifestations of FFD when applied in action**.
Together, they show that frontiers arise not only as search stagnation, but also as collapse of guidance and canonicity.

The contribution of this paper is therefore twofold: it provides new structural impossibility theorems, and it demonstrates that **FFD functions as a generative tool for identifying classes of necessary failure modes** under fixed observation.
Implications for rational search control and abstraction change are discussed, preparing the ground for subsequent work.

Keywords: Reasoning systems; Search control; Observational granularity; Search stagnation; #SAT; Canonical selection; Proof search; Structural limits of computation; General topology; Ramsey theory; Petri nets

---

1. Introduction — From Framework to Action

In many areas of theoretical computer science and mathematics, one encounters persistent failures that are difficult to classify.
Search procedures stagnate despite continued internal activity, counting problems resist stabilization from partial information, and existence results fail to admit canonical or coherent choices.
Such phenomena are often treated as unrelated pathologies, explained case by case through domain-specific arguments or attributed to limitations of particular methods.

Frontier Field-Dynamics (FFD), introduced in a companion foundational paper, provides a structural perspective on these failures.
FFD analyzes reasoning and search processes under **fixed observational granularity**, explicitly separating internal evolution from observable informational progress. Within this framework, frontiers are identified as regions where extensive internal change produces no gain at the level of observation.

The purpose of the present paper is to move **from framework to action**.
Rather than further developing the formal apparatus of FFD, we ask a different question:

What kinds of results are structurally forced when FFD is applied to concrete mathematical settings?

Our main thesis is that the answer is remarkably rigid.
Applying FFD does not lead to a heterogeneous collection of phenomena, but consistently produces **structural impossibility results**. These results are not artifacts of specific models, encodings, or heuristics; they arise inevitably once observational granularity is fixed and coherence requirements are imposed.

In this paper, we analyze two such impossibility mechanisms.

The first concerns **finite-grain stabilization in counting problems**.
We show that the counting function of the canonical problem #SAT does not factor through any finite-grain observational scheme, even pointwise. At every finite level of information, extensions of the same partial description can yield different solution counts. This establishes a model-independent oscillation phenomenon that prevents localization of global counts from finite observations.

The second mechanism concerns **canonical selection under natural coherence constraints**.
We prove a general no-go theorem showing that no non-trivial selection rule can be simultaneously intrinsic, compatible with loss of information, and stable under extension. This obstruction is purely structural and independent of quantitative bounds or domain-specific constructions. As immediate corollaries, it explains familiar non-canonicity phenomena in general topology, Ramsey-type settings, and concurrent systems such as Petri nets.

These two results are mathematically independent and arise in different domains.
Their unifying feature is not their subject matter, but the fact that **they are structurally inevitable under FFD**. In the first case, the frontier manifests as persistent informational oscillation; in the second, as collapse of guidance and canonicity. Together, they show that frontiers are not limited to search stagnation, but also govern the failure of selection, stabilization, and control.

It is important to stress what this paper does not aim to do.
We do not propose new algorithms, heuristics, or optimization techniques. We do not claim complexity-theoretic separations, nor do we attempt to resolve open problems by constructive means. The results presented here are not offered as isolated discoveries, but as **structural consequences of applying FFD in action**.

The contribution of this paper is therefore twofold.
First, it establishes new impossibility theorems of independent mathematical interest.
Second, and more importantly, it demonstrates that FFD functions as a **generative framework**: when applied to concrete settings, it restricts the space of possible outcomes to specific classes of failure modes. This perspective provides a unifying explanation for why certain problems resist stabilization, canonization, or control under finite observation.

The paper is organized as follows.
Section 2 briefly recalls the minimal aspects of Frontier Field-Dynamics needed for the present analysis.
Section 3 clarifies what it means to “apply” FFD in a mathematical context.
Sections 4 and 5 present two distinct frontier mechanisms together with their associated impossibility results.
Section 6 discusses structural corollaries across several domains.
Section 7 explains why these results are not surprising under FFD and outlines their implications for rational search control, preparing the ground for subsequent work.
Complete formal proofs of the main impossibility results are provided in Appendix A and Appendix B.

---

2. Minimal Recall: What Frontier Field-Dynamics Detects

Frontier Field-Dynamics (FFD) is a structural framework for analyzing reasoning and search processes under **fixed observational granularity**.
Rather than modeling specific algorithms or inference rules, FFD abstracts reasoning as a dynamical evolution of internal states coupled with an explicit notion of observation.

In the FFD setting, a reasoning process is described by:

- an internal state space, representing the evolving configurations of the system;
- an observational projection, capturing which distinctions are accessible at a given level of abstraction.

The observational projection induces a notion of **observable information**.
Internal evolution is said to produce progress only insofar as it yields new observable distinctions. Internal change that does not alter observable information is treated as epistemically inert.

A central concept of FFD is that of a **frontier**.
Informally, a frontier is a region of the state space in which internal evolution persists but observable information fails to improve. More precisely, a frontier arises whenever, under a fixed observational regime, the induced dynamics on observations either stabilizes without resolving the relevant task or fails to stabilize altogether.

Two features of FFD are essential for the present work.

First, FFD is **structural rather than algorithmic**.
It does not depend on computational resources, heuristics, or complexity assumptions. Frontiers are defined in terms of informational accessibility, not efficiency or runtime.

Second, FFD is **agnostic to domain-specific formalism**.
The same notions apply to proof search, counting problems, selection procedures, and other reasoning tasks, provided that a notion of internal evolution and observational granularity is specified.

The role of FFD is therefore not to predict specific outcomes, but to **restrict the space of admissible behaviors** under fixed observation. Once the observational regime is fixed, only certain structural patterns can arise: stabilization, oscillation, or failure of guidance. Results incompatible with these patterns are structurally excluded.

This minimal recall suffices for the analysis that follows.
The full formal development of Frontier Field-Dynamics is given in the companion foundational paper; here we rely only on the concepts outlined above.

---

3. What It Means to Apply Frontier Field-Dynamics

Applying Frontier Field-Dynamics (FFD) does not mean using it as an algorithmic technique, a heuristic improvement, or a method for constructing solutions.
FFD is not applied to a problem in order to solve it, but to the space of possibilities in which reasoning unfolds, in order to determine what kinds of outcomes are structurally admissible under fixed observational granularity.

This distinction is crucial. In most mathematical and computational contexts, “application” suggests operational use: implementing a procedure, optimizing a strategy, or improving performance. FFD operates at a different level. It analyzes the **conditions under which progress, stabilization, or guidance are even possible**, independently of how much computational effort is expended.

Concretely, applying FFD consists of the following steps.

First, one fixes a notion of **state** and a corresponding **observational projection**, representing what information is operationally accessible at a given level of abstraction. The choice of projection

determines the observational granularity and is treated as fixed throughout the analysis. No assumptions are made about optimality, completeness, or efficiency of this choice.

Second, one considers the induced dynamics on observations: how internal evolution projects onto observable change. Informational progress is defined strictly in terms of changes at the observational level, not in terms of internal activity or syntactic transformation.

Third, one analyzes the **structural behavior** of this induced dynamics. In particular, one asks whether observable information stabilizes, oscillates, or fails to admit coherent selection or guidance. Regions in which internal evolution persists without observable progress are identified as frontiers.

Importantly, this analysis is **agnostic to domain-specific details**. FFD does not depend on the nature of the underlying objects (formulas, configurations, executions), nor on the computational model used to explore them. What matters is only the interaction between internal evolution and fixed observational granularity.

From this perspective, the role of FFD is not predictive in the usual sense.
FFD does not predict which specific theorem will hold in a given domain. Rather, it **restricts the space of possible outcomes**. Once the observational regime is fixed, only certain classes of behavior can occur: stabilization, persistent oscillation, collapse of guidance, or non-canonicity. Results that fall outside these classes are structurally excluded.

The results presented in the following sections should be read in this light.
They are not isolated mathematical curiosities, nor counterexamples tailored to particular settings. Instead, they are **structurally forced manifestations** of applying FFD to concrete problems. The domains differ, but the underlying mechanisms do not.

In this sense, the proofs that follow play a dual role.
They establish new impossibility theorems of independent interest, while simultaneously serving as **evidence that FFD, when put in action, consistently generates results of a specific structural type**. The value of these results lies not only in their statements, but in the fact that their form and content are dictated by the same underlying framework.

With this interpretation in place, we now turn to the first frontier mechanism, concerning the impossibility of finite-grain stabilization in counting problems.

---

4. Frontier Mechanism I — Finite-Grain Non-Factorization of #SAT

4.1 Formal setting: finite-grain observation of extensible instances

We consider instances of #SAT represented via an **extensible encoding** into finite strings, allowing incremental construction by extension.
Finite observational access is modeled by **finite-grain observation schemes**, i.e. families of equivalence relations with finitely many classes at each level and monotone refinement.

Given such a scheme, two instances are observationally indistinguishable at level k if they belong to the same equivalence class of the k-th observation.

The question is whether the counting function

$$\#\mathrm{SAT}: \mathrm{Instances} \to \mathbb{N}$$

can **stabilize** with respect to such finite-grain observations.

4.2 Local oscillation lemma

We first isolate the key structural phenomenon.

Lemma 4.1 (Local Oscillation).
Let π be any finite-grain observation scheme on extensible #SAT instances.
Then for every observation level $k$and every observational class Cat level k, there exist instances $x, y \in C$such that

$$\#\mathrm{SAT}(x) \neq \#\mathrm{SAT}(y).$$

In particular, no finite observational class determines the value of the counting function.

Proof sketch.
Fix a finite observational class C, corresponding to a finite prefix or finite description of an instance.
By extensibility, this partial description admits multiple completions. One can construct two such completions that differ only on fresh variables, yielding formulas with different numbers of satisfying assignments (e.g. satisfiable vs. unsatisfiable extensions).
Since these extensions remain indistinguishable at level k, the counting function oscillates within C.
■

4.3 Main theorem: finite-grain non-factorization

The oscillation phenomenon rules out any form of stabilization.

Theorem 4.2 (Finite-Grain Non-Factorization of #SAT).
The counting function #SAT does not factor through any finite-grain observation scheme, even pointwise.
Equivalently, there exists no finite observational level—possibly depending on the instance—at which the value of #SAT becomes determined.

Proof sketch.
Assume, for contradiction, that #SAT stabilizes with respect to some finite-grain scheme.
Then for each instance there exists a finite observational level whose equivalence class determines the value of #SAT.
This contradicts Lemma 4.1, which shows that oscillation persists within every finite observational class. ■

A complete formal proof is given in Appendix A.

4.4 Interpretation as a frontier under FFD

From the perspective of Frontier Field-Dynamics, Theorem 4.2 identifies a **frontier induced by scale**.

Internally, instances evolve freely by extension.
Observationally, however, finite-grain access never suffices to stabilize the global count. The reasoning process may explore indefinitely, yet no observable informational progress is possible at the fixed granularity.

This is precisely the structural signature of a frontier:
persistent internal evolution without observable stabilization, and impossibility of progress without a change of observational regime.

4.5 Structural status of the result

The non-factorization of #SAT is not an algorithmic limitation.
It does not depend on computational bounds, reductions, or complexity-theoretic assumptions.
Instead, it reflects an intrinsic incompatibility between:

- incremental construction,
- finite observational granularity,
- and global counting quantities.

In this sense, the theorem should be read as a **necessary outcome of applying FFD** to counting problems, rather than as a problem-specific anomaly.

---

5. Frontier Mechanism II — Structural Impossibility of Canonical Selection

5.1 Selection structures

We introduce a minimal abstract setting sufficient to formalize canonical selection problems.

Definition 5.1 (Selection Structure).
A selection structure is a tuple

$$\mathcal{S} = (\mathrm{O},\ \sim,\ \preceq_{\mathrm{r}},\ \preceq_{\mathrm{e}},\ \mathrm{Sel})$$

where:

- O is a class of objects;
- $\sim$ is an equivalence relation on O (structural equivalence);
- $\preceq_{\mathrm{r}}$ is a preorder on O (reduction);

- $\preceq_e$ is a preorder on $\mathcal{O}$ (extension);
- for each $O \in \mathcal{O}$, $\mathrm{Sel}(O)$ is a nonempty set of admissible regimes associated with O.

No compatibility between $\preceq_r$ and $\preceq_e$ is assumed beyond their intended interpretation as loss and gain of information.

5.2 Coherent selection rules

We now formalize the notion of canonicity via coherence requirements.

Definition 5.2 (Coherent Selection Rule).
A selection rule is a map

$$s: \mathcal{O} \longrightarrow \bigcup_{O \in \mathcal{O}} \mathrm{Sel}\,(O) \text{ with } s(O) \in \mathrm{Sel}(O).$$

The rule s is coherent if it satisfies the following conditions:

1. **Invariance**:
   If $O_1 \sim O_2$, then $s(O_1) \sim s(O_2)$.
2. **Reduction compatibility**:
   If $O_1 \preceq_r O_2$, then the reduction of $s(O_2)$ is equivalent to $s(O_1)$.
3. **Extension stability**:
   If $O_1 \preceq_e O_2$, then $s(O_1)$ remains admissible for O_2.

These conditions express the requirement that canonical choices depend only on intrinsic structure, behave consistently under loss of information, and persist under refinement.

5.3 Main impossibility theorem

We now state the central result.

Theorem 5.3 (Structural Impossibility of Canonical Selection).
Let $\mathcal{S}$ be any selection structure.
There exists no coherent selection rule satisfying the conditions of Definition 5.2 unless the rule is trivial, i.e. fails to distinguish objects in a structurally meaningful way.

Proof sketch.
Assume a non-trivial coherent selection rule s exists.

Invariance forces s to ignore all distinctions not preserved under $\sim$.
Reduction compatibility forces s to be insensitive to information that may be collapsed.
Extension stability forces s to depend only on features that persist under arbitrary refinement.

Any feature capable of inducing a non-trivial distinction must fail at least one of these constraints: it is either destroyed by reduction or altered by extension. Hence no such feature can be used by s.

Therefore, coherence forces s to collapse to a degenerate assignment, contradicting non-triviality. ■

A complete formal proof is given in Appendix B.

5.4 Interpretation as a frontier under FFD

Within Frontier Field-Dynamics, Theorem 5.3 identifies a **frontier of guidance**.

Internal evolution may generate multiple admissible regimes for a given object.
However, under fixed observational granularity and coherence constraints, no rule can canonically privilege one regime over another. Additional refinement or exploration does not reduce ambiguity; it preserves it.

In contrast to the oscillatory frontier of Section 4, this frontier manifests as **structural non-selectability**.
Progress is possible, but guidance collapses: the reasoning process cannot rationally commit to a distinguished path without changing the observational or structural regime.

5.5 Summary of the second frontier mechanism

The second frontier mechanism can be summarized as follows:

Under natural coherence constraints, canonical selection is structurally impossible.

This impossibility is independent of domain, model, or computational resources.

Together with the finite-grain oscillation mechanism of Section 4, it completes the classification of frontier behaviors forced by applying FFD under fixed observation.

---

6. Structural Corollaries Across Domains

The impossibility theorems of Sections 4 and 5 describe two abstract frontier mechanisms:
(i) failure of stabilization under finite observation, and
(ii) impossibility of coherent canonical selection.

We now show that several well-known non-canonicity phenomena across different domains arise as **direct corollaries** of these mechanisms.
No new structural assumptions are introduced.

6.1 General Topology — Canonical Bases

In general topology, every space admits bases of minimal cardinality.
However, such bases are highly non-unique.

Corollary 6.1 (Non-Existence of Canonical Minimal Bases).
There exists no assignment that associates to each topological space a minimal base in a way that is:

1. invariant under homeomorphism,
2. compatible with quotients and restrictions,
3. stable under refinement or extension of the space.

Any such assignment must be trivial.

Justification.
Topological spaces, together with homeomorphisms, quotients, and refinements, form a selection structure in the sense of Definition 5.1.
The coherence requirements correspond exactly to those of Definition 5.2.
The claim follows directly from Theorem 5.3. ■

6.2 Ramsey Theory — Witness Selection

Ramsey-type theorems guarantee the existence of witnesses (e.g. monochromatic substructures) in sufficiently large configurations, but do not provide canonical choices.

Corollary 6.2 (Impossibility of Equivariant Witness Selection).
In Ramsey-type existence settings, there exists no non-trivial witness selection rule that is:

1. invariant under symmetries of the configuration,
2. compatible with restriction to subconfigurations,
3. stable under extension of the ambient structure.

Justification.
Configurations, together with symmetry equivalence, restriction, and extension, define a selection structure.
Any equivariant witness selector satisfying the above properties would constitute a coherent selection rule, contradicting Theorem 5.3. ■

6.3 Concurrent Systems — Petri Nets

Petri nets admit multiple executions that are equivalent from a structural standpoint.

Corollary 6.3 (Non-Canonicity of Executions in Petri Nets).
There exists no canonical execution selector for Petri nets that is:

1. invariant under net isomorphism,
2. compatible with projection to subnets,
3. stable under extension by independent components.

Justification.
Executions of Petri nets form admissible regimes associated with a net.
Isomorphism, projection, and extension induce equivalence, reduction, and extension relations, respectively.
A canonical execution selector satisfying the above properties would be a coherent selection rule, contradicting Theorem 5.3. ■

6.4 Structural role of the corollaries

Corollaries 6.1–6.3 are not domain-specific anomalies.
They are instances of the same abstract obstruction identified in Section 5.

In each case:

- existence of admissible regimes is guaranteed,
- multiple regimes coexist,
- coherence requirements rule out canonical selection.

These corollaries therefore provide independent confirmation that the second frontier mechanism captures a **generic structural phenomenon**, rather than an artifact of a particular formalism.

---

7. Why These Results Are Structurally Forced

The results of Sections 4 and 5 may appear, at first glance, as independent impossibility theorems arising in different contexts.
One concerns stabilization of global counts under finite observation; the other concerns the impossibility of coherent canonical selection.
From the perspective of Frontier Field-Dynamics, however, **their appearance is not contingent**.

Once the observational regime is fixed, FFD restricts the space of admissible outcomes.
Reasoning processes can only exhibit a limited number of structural behaviors at a given level of granularity. In particular, observable information may either stabilize, oscillate, or fail to admit coherent guidance. Outcomes outside these classes are structurally excluded.

The first frontier mechanism identifies the case in which **observable quantities fail to stabilize**.
Despite unrestricted internal evolution, finite-grain observation cannot localize global information.
Oscillation is therefore unavoidable. The non-factorization of #SAT is not a pathology of a specific problem, but the canonical manifestation of this regime.

The second frontier mechanism identifies a complementary failure mode.
Here, observable information may be stable, and multiple admissible regimes may exist, but **no coherent criterion can distinguish one as canonical**. Guidance collapses structurally, even though existence is guaranteed. The impossibility of canonical selection captures this regime in its purest form.

These two mechanisms exhaust the ways in which frontiers arise under fixed observation.
Either progress fails because observable information never stabilizes, or it fails because stabilization does not induce guidance. In both cases, further internal exploration, refinement, or optimization cannot resolve the obstruction as long as the observational regime remains unchanged.

From this perspective, the results of this paper are **forced**.
Given finite observational granularity and natural coherence requirements, impossibility theorems of the kind presented here are not exceptional outcomes; they are the only admissible ones. What would require explanation is not their existence, but their absence.

This observation clarifies the role of Frontier Field-Dynamics.
FFD does not predict specific theorems in advance, nor does it subsume domain-specific arguments. Instead, it functions as a **structural filter**: it constrains the form that any result can take once the observational assumptions are fixed.

In particular, FFD explains why attempts to overcome the impossibilities identified in Sections 4 and 5 by purely local means—stronger heuristics, refined constructions, or extended search—are necessarily ineffective. Such attempts do not address the source of the obstruction, which lies at the level of observation and coherence, not at the level of exploration.

The significance of this perspective extends beyond the individual results presented here.
It suggests that impossibility theorems arising in different domains should not be treated as isolated barriers, but as indicators of underlying frontier mechanisms. Recognizing which mechanism is at work is essential for determining whether continued effort is rational, or whether a change of observational regime is required in principle.

This insight prepares the ground for the next step.
If frontiers are structural and unavoidable under fixed observation, then reasoning systems require criteria for **rational control**: criteria for stopping, switching, or changing abstraction when progress is no longer possible. The development of such criteria lies beyond the scope of the present paper and is addressed in subsequent work.

---

8. Conclusion

This paper has examined what happens when Frontier Field-Dynamics (FFD) is applied to concrete mathematical settings.
Rather than yielding isolated or domain-specific phenomena, FFD consistently forces **structural impossibility results** once observational granularity and coherence constraints are fixed.

We identified two distinct frontier mechanisms.
The first manifests as **failure of stabilization**: global quantities cannot be determined from finite-grain observation, leading to unavoidable oscillation.
The second manifests as **collapse of canonical guidance**: although admissible regimes exist, no coherent criterion can select one as canonical.

These mechanisms arise independently of computational resources, algorithms, or formal encodings.

They reflect intrinsic structural constraints on reasoning under fixed observation, not limitations of particular methods. In this sense, the impossibility results presented here are not accidental; they are **necessary outcomes** of the underlying framework.

By formalizing these mechanisms and exhibiting their recurrence across domains, the paper clarifies why certain problems resist stabilization, canonization, or control despite extensive internal exploration.
It also explains why purely local refinements—stronger heuristics, deeper search, or additional constraints—cannot overcome these obstacles without a change of observational regime.

The role of FFD is therefore not to provide solutions, but to **delineate the boundary of what is structurally achievable** under given observational assumptions.
Recognizing when a reasoning process has entered a frontier is essential for deciding whether continued effort is rational or whether a change of abstraction is required in principle.

The implications of this perspective for **search control and rational decision-making** are substantial.
Developing explicit criteria for stopping, switching, or changing observational regimes lies beyond the scope of the present work and is addressed in subsequent research.

Appendix A

Finite-Grain Non-Factorization of #SAT

This appendix provides the complete formal development underlying **Theorem 4.2**.
All definitions, lemmas, and proofs are given in full, making the result self-contained and definitive.

A.1 Extensible encodings and prefix space

We work in the space $X := \{0,1\}^*$ of all finite binary strings, endowed with the prefix order $\preceq$.
For $p \in X$, define the basic neighborhood

$$N(p) := \{q \in X : q \succeq p\}.$$

Let $\mathcal{F}$ denote the set of Boolean formulas in conjunctive normal form (CNF).

Definition A.1 (Extensible Encoding).
An encoding

$$\mathrm{enc} : \mathcal{F} \to X$$

is extensible if for every prefix $p \in X$ that is a prefix of at least one valid encoding, and for every CNF formula F, there exists a string $q \succeq p$ such that:

1. $q = \mathrm{enc}(F')$ for some CNF formula F';
2. F' is obtained from F by adding clauses involving only variables not occurring in F.

This property ensures that no finite prefix can determine the global structure of an instance: any partial description admits arbitrarily many completions by extension.

A.2 The counting function

Given an extensible encoding, define the counting function

$$\#: X \longrightarrow \mathbb{N}$$

by letting $\#(x)$ be the number of satisfying assignments of the CNF formula decoded from x.

All results below are invariant under the chosen encoding, **provided extensibility holds**.

A.3 Finite-grain observation schemes

Definition A.2 (Finite-Grain Observation Scheme).
A finite-grain observation scheme is a family $\{\sim_k\}_{k\in\mathbb{N}}$ of equivalence relations on X such that:

1. for each k, the quotient $\frac{X}{\sim_k}$ has finitely many equivalence classes;
2. if $k \leq \ell$, then $\sim_\ell$ refines $\sim_k$;
3. **each equivalence class of $\sim_k$ is a union of prefix neighborhoods**.

Condition (3) expresses that finite-grain observations respect the prefix structure of extensible encodings and capture only finite initial information.

A.4 Pointwise stabilization

We formalize the notion of stabilization used in the main result.

Definition A.3 (Pointwise Stabilization).
The counting function $\#$ stabilizes pointwise with respect to a finite-grain observation scheme $\{\sim_k\}$ if for every $x \in X$ there exists $k \in \mathbb{N}$ such that $\#$ is constant on the equivalence class $[x]_{\sim_k}$.

A.5 Local oscillation lemma

We now establish the key structural property.

Lemma A.4 (Local Oscillation).
Let enc be an extensible encoding.
For every prefix $p \in X$, there exist extensions $y, z \succeq p$ such that

$$\#(y) \neq \#(z).$$

Proof.
Fix an arbitrary prefix p.
By extensibility, there exists at least one completion $y \succeq p$ decoding to a CNF formula F.

Using the same prefix p, extensibility guarantees the existence of another completion $z \succeq p$ decoding to a formula F' obtained by appending clauses on fresh variables.
Choose such an extension z so that F' is unsatisfiable, for instance by adding the clauses (v) and $(\neg v)$ for a new variable v. Then $\#(z) = 0$.

On the other hand, choose y so that the resulting formula remains satisfiable (for example by appending no clauses, or only tautological ones), yielding $\#(y) \geq 1$.

Thus $\#(y) \neq \#(z)$, and both strings extend the same prefix p. ■

A.6 Finite-grain non-factorization theorem

We now derive the main result.

Theorem A.5 (Finite-Grain Non-Factorization of #SAT).
Let enc be an extensible encoding of #SAT instances, and let $\{\sim_k\}_{k\in\mathbb{N}}$ be any finite-grain observation scheme on X.
Then the counting function

$$\#: X \longrightarrow \mathbb{N}$$

does not stabilize pointwise with respect to $\{\sim_k\}$.
Consequently, #does not factor through any finite-grain observation scheme.

Proof.
Fix an arbitrary observation scheme $\{\sim_k\}$.
Let k be arbitrary and let C be any equivalence class of $\sim_k$.

By Definition A.2(3), there exists a prefix p such that infinitely many elements of C extend p.
By Lemma A.4, there exist two extensions $y, z \succeq p$ such that $\#(y) \neq \#(z)$.
Since both y and z lie in C, the function # is not constant on C.

As k and C were arbitrary, no pointwise stabilization can occur.
Hence # does not admit any finite-grain factorization. ■

A.7 Structural interpretation

Theorem A.5 shows that the value of #SAT is **intrinsically non-local** under extensible encodings. No finite observational regime—regardless of how it is organized or refined—can determine the global count.

This impossibility is structural rather than algorithmic.
It does not rely on computational bounds, reductions, or complexity-theoretic assumptions, but solely on the interaction between incremental construction, finite observation, and global counting.

Appendix B

Structural Impossibility of Canonical Selection under Natural Coherence Constraints

This appendix provides the complete formal development underlying **Theorem 5.3**.
All definitions, lemmas, and proofs are given explicitly, making the impossibility result self-contained and definitive.

B.1 Selection structures

We formalize the abstract setting in which canonical selection problems arise.

Definition B.1 (Selection Structure).
A selection structure is a tuple

$$\mathcal{S} = (\mathcal{O},\ \sim,\ \preceq_{\mathrm{r}},\ \preceq_{\mathrm{e}},\ \mathrm{Sel})$$

where:

1. $\mathcal{O}$ is a class of objects;
2. $\sim$ is an equivalence relation on $\mathcal{O}$ (structural equivalence);
3. $\preceq_{\mathrm{r}}$ is a preorder on $\mathcal{O}$ (reduction);
4. $\preceq_{\mathrm{e}}$ is a preorder on $\mathcal{O}$ (extension);
5. for each $\mathrm{O} \in \mathcal{O}$, $\mathrm{Sel}(\mathrm{O})$ is a nonempty set of admissible regimes associated with O.

No algebraic, topological, or computational structure is assumed beyond these relations.

B.2 Selection rules and coherence

Definition B.2 (Selection Rule).
A selection rule is a map

$$\mathrm{s}: \mathcal{O} \longrightarrow \bigcup_{\mathrm{O} \in \mathcal{O}} \mathrm{Sel}\,(\mathrm{O}), \mathrm{s}(\mathrm{O}) \in \mathrm{Sel}(\mathrm{O}).$$

We formalize canonicity via coherence requirements.

Definition B.3 (Coherence Conditions).
A selection rule s is coherent if it satisfies:

1. **Invariance**
   If $\mathrm{O}_1 \sim \mathrm{O}_2$, then $\mathrm{s}(\mathrm{O}_1) \sim \mathrm{s}(\mathrm{O}_2)$.

2. **Reduction compatibility**
   If $O_1 \preceq_r O_2$, then the reduction of $s(O_2)$ is equivalent to $s(O_1)$.
3. **Extension stability**
   If $O_1 \preceq_e O_2$, then $s(O_1)$ remains admissible for O_2.

These conditions express the minimal requirements for a choice to be canonical: intrinsic dependence, coherence under loss of information, and persistence under refinement.

B.3 Trivial and non-trivial selection

The impossibility result hinges on a precise notion of triviality.

Definition B.4 (Degenerate Selection Rule).
A selection rule s is degenerate if it is invariant under both reduction and extension, i.e. for all $O_1, O_2 \in \mathcal{O}$,

$$\forall\, O_1,\, O_2, \qquad O_1 \preceq_r O_2 \;\Rightarrow\; s(O_1) \sim s(O_2), \text{and } O_1 \preceq_e O_2 \;\Rightarrow\; s(O_1) \sim s(O_2).$$

Equivalently, a degenerate rule fails to distinguish objects in a structurally meaningful way.

A selection rule is non-trivial if it is not degenerate.

B.4 Core incompatibility lemma

We now isolate the key structural obstruction.

Lemma B.5 (Incompatibility of Non-Trivial Distinctions).
Let s be a coherent selection rule on a selection structure $\mathcal{S}$.
Then any structural distinction used by s must be invariant under both reduction and extension.

Proof.
By reduction compatibility, any feature of O on which $s(O)$ depends must persist under all reductions $\preceq_r$; otherwise, the selected regime would not be compatible with loss of information.

By extension stability, any such feature must also persist under all extensions $\preceq_e$; otherwise, the selected regime would cease to be admissible upon refinement.

Therefore, any distinction that influences the selection must be invariant under both reduction and extension. ■

B.5 Main impossibility theorem

We can now state and prove the central result.

Theorem B.6 (Structural Impossibility of Canonical Selection).
There exists no coherent non-trivial selection rule on any selection structure.

Proof.
Assume that a coherent non-trivial selection rule s exists.

By Lemma B.5, every distinction used by s must be invariant under both reduction and extension.
By Definition B.4, this implies that s is degenerate.

This contradicts the assumption that s is non-trivial.
Hence no coherent non-trivial selection rule exists. ■

B.6 Relation to domain-specific corollaries

Theorem B.6 is purely structural and domain-independent.
Its instantiations in general topology, Ramsey-type settings, and concurrent systems are discussed in **Section 6** of the main paper.

Each of those settings admits a natural selection structure satisfying Definitions B.1–B.3.
The absence of canonical bases, equivariant witnesses, or distinguished executions follows directly from Theorem B.6.

B.7 Structural interpretation

Theorem B.6 identifies a fundamental obstruction to canonical guidance.
Even when admissible regimes exist, coherence under equivalence, reduction, and extension forces selection to collapse.

This impossibility is not algorithmic, quantitative, or domain-specific.
It arises solely from the structural requirements imposed on any notion of canonical choice.

References

De Palma, F. *Frontier Field-Dynamics: A Structural Framework for Reasoning under Fixed Observation*, 2026.

Cook, S. A. *The Complexity of Theorem-Proving Procedures*, STOC, 1971.

Garey, M. R., Johnson, D. S. *Computers and Intractability*, Freeman, 1979.

Ramsey, F. P. *On a Problem of Formal Logic*, Proc. London Math. Soc., 1930.

Winskel, G. *Event Structures*, Springer, 1987.

Bourbaki, N. *General Topology*, Springer.